

LC 2615: Sum of Distances

Target Companies: BNY, Adobe

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Date: September 19, 2026

Problem Breakdown

basically

Given an array $[1, 3, 1, 1, 2]$, then for all i, j where $arr[i] == arr[j]$, need to return an array where $arr[i]$ is sum of all such $|i - j|$.

Array:

1	3	1	1	2
0	1	2	3	4

Left Contribution Logic

So for the element 1, the indices are $[0, 2, 3]$.

we need $(3-0) + (2-0)$ for index 0

or

$abs(0-3) + abs(0-2)$

$(0 * 2) + (2 + 3)$

i.e

~~$(idx * count) + \text{sum of already processed}$~~ (Crossed out in notes)

Eg for index 3 ka one

$(3-2) + (3-0)$

$(3 * 2) - (2 + 0)$

Formula Derived: $[idx * count - sum]$

 Handwritten Note 1

Fig 1: Original notes for Left Contribution.

Right Contribution Logic

however need to consider right side too

Eg take one which was index 2 then

Indices for '1':

0	2	3
idx:0	idx:2	idx:3

left contri = $(2 * 1) - 0 = 2$

right contri = $(2 - 3)$ i.e $(3 - 2) = 1$

can be represented as $(\text{rightsum}) - (x * \text{cnt})$

So:

left contri = $(x * \text{leftcount} - \text{leftsum})$

right contri = $(\text{rightsum} - x * \text{rightcnt})$



Handwritten Note 2

Fig 2: Original notes for Right Contribution and final formulas.

C++ Implementation

Below is the optimal code structure following the derived prefix-sum and grouping logic. Code is preserved exactly as authored.

```
class Solution { public: vector<long long> distance(vector<int>& nums) { int n =
```